

KNN

Classification (predicting classes)

e.g Predicting movie Genre

IMDb Rating	Duration	Genre
8.0 (Mission Impossible)	160	Action
6.2 (Predator 2)	170	Action
7.2 (Rocky and Bullwinkle)	168	Comedy
8.2 (OMG 2)	155	Comedy

Now predict the genre of "Barbie" movie with IMDb rating 7.4 and duration 114

Step 1: Calculation Distance (Euclidean distance)

Calculate the Euclidean distance b/w the new movie and each movie in the dataset

$$d(p, q) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\begin{aligned} x_1 &= 7.4 \\ y_1 &= 114 \end{aligned}$$

$$\text{Distance to } (8.0, 160) = \sqrt{((7.4 - 8.0)^2 + (114 - 160)^2)} = \approx 46.00$$

$$\text{Distance to } (6.2, 160) = \sqrt{((7.4 - 6.2)^2 + (114 - 170)^2)} = \approx 56.01$$

$$\text{Distance to } (7.2, 168) = \sqrt{((7.4 - 7.2)^2 + (114 - 168)^2)} = \approx 54.00$$

$$\text{Distance to } (8.2, 155) = \sqrt{((7.4 - 8.2)^2 + (114 - 155)^2)} = \approx 41.00$$

Step 2:

Select k Nearest Neighbors

Suppose $k=1$ (Neighbors)

Small distance 41.00

Best value (5)

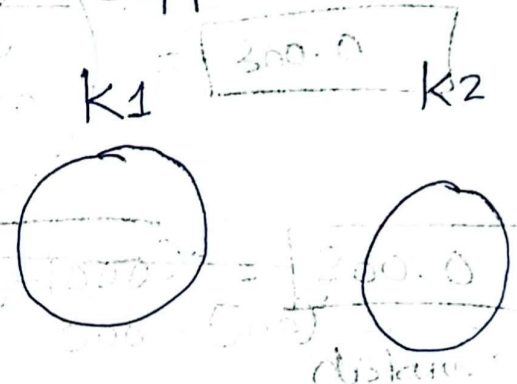
Step 3: Majority voting (A, C, C)
classification

Barbie move "C"

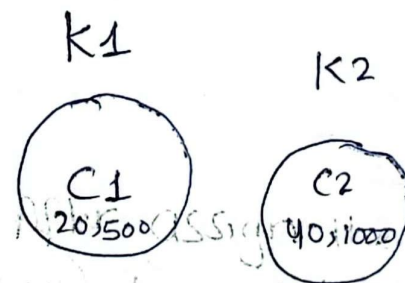
k-mean clustering algorithm

Sri.No	age	amount
C1	20	500
C2	40	1000
C3	30	800
C4	18	300
C5	28	1200
C6	35	1400
C7	45	1800

Step 1: Suppose two cluster



Step 2: Centroid value (Random Select)



Step 3:

C3 = Nearest distance check it

By using Euclidean Distance

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

obj
Centroid

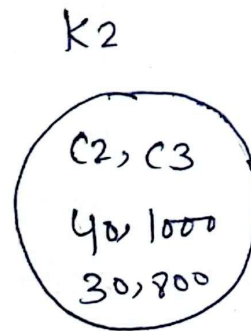
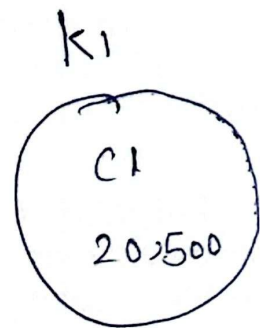
Step 4:

$$k_1 \rightarrow C_3 = \sqrt{(30 - 20)^2 + (800 - 500)^2}$$

$$C_3 = \sqrt{(10)^2 + (300)^2} = \boxed{300.0}$$

$$k_2 \rightarrow C_3 = \sqrt{(30 - 40)^2 + (800 - 1000)^2} = \boxed{200.0}$$

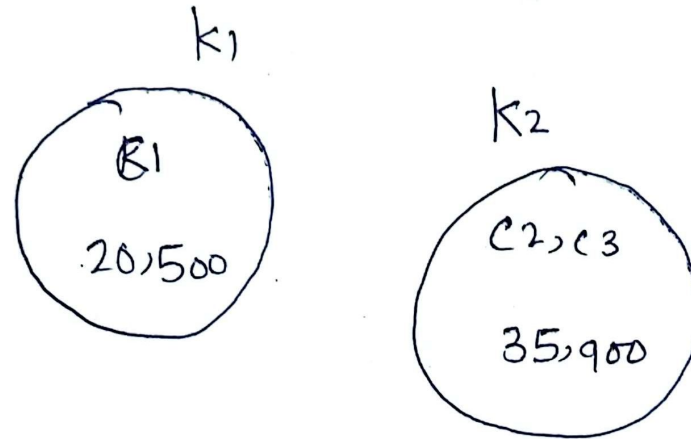
~~100 - 500~~ distance small



After assign the value
must update the centroid
value.

$$\frac{800 + 1000}{2} = \boxed{900} \quad ; \quad \frac{40 + 30}{2} = \boxed{35}$$

Step 5:



Step 6:

$$C_4 = \sqrt{(18-20)^2 + (300-500)^2} = \text{small value}$$

$$C_4 = \sqrt{(18-35)^2 + (300-900)^2} =$$

